Loss Functions for Oriented Bounding Box Detection

Mathematical Motivation, ProbIoU, and YOLO-OBB

CS 539 — Computer Vision Lecture Notes

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Abstract

We derive a geometry-aware loss for rotated bounding boxes by representing each oriented rectangle through the mean and covariance of a two-dimensional Gaussian. We then connect Bhattacharyya distance, probabilistic IoU (ProbIoU), classification loss, and distribution focal loss (DFL) to the practical YOLO-OBB training objective. The central theme is *representation invariance*: distinct angle parameters may describe exactly the same rectangle.

1 The geometric estimation problem

An axis-aligned rectangle is represented as $B = (c_x, c_y, w, h)$. An oriented bounding box (OBB) adds an angle:

$$\boxed{B = (c_x, c_y, w, h, \alpha), \qquad w, h > 0.} \tag{1}$$

The objective of a detection network $f_{\boldsymbol{\theta}}$ is to map an image x to candidate boxes, object scores, and classes. Here $\boldsymbol{\theta}$ denotes *network parameters*; α denotes the rotation angle of a box. A training objective has the schematic form

$$\boldsymbol{\theta}^* \approx \arg\min_{\boldsymbol{\theta}} \sum_{j=1}^{N} \mathcal{L}(f_{\boldsymbol{\theta}}(x_j), y_j), \tag{2}$$

where the actual detector also needs a procedure that assigns predictions to ground-truth objects.

2 Why naive coordinate MSE is problematic

A tempting approach is $\sum_k (b_k - \hat{b}_k)^2$ over the five box parameters. It is not geometrically invariant. For example,

$$\boxed{(c_x, c_y, w, h, \alpha) \sim (c_x, c_y, w, h, \alpha + \pi)} \tag{3}$$

represents exactly the same rectangle. There is a second equivalence,

$$\boxed{(c_x, c_y, w, h, \alpha) \sim (c_x, c_y, h, w, \alpha + \pi/2).} \tag{4}$$

Yet coordinate-wise MSE penalizes both. Moreover, position errors have pixel units, angle errors have radian units, and a 5° angular error displaces the ends of a long, thin register much more than those of a nearly square box. The useful loss should account for the *whole geometry* rather than arbitrary parameter conventions.

One alternative is polygon IoU,

$$\mathrm{IoU}_{\mathrm{rot}}(B, \hat{B}) = \frac{|B \cap \hat{B}|}{|B \cup \hat{B}|}. \tag{5}$$

However, explicitly differentiating polygon intersection under all geometric configurations is inconvenient. Gaussian-based rotated-box losses provide a smooth, geometry-aware surrogate.

3 Representing a rectangle as a Gaussian

Identify an OBB with a Gaussian $\mathcal{N}(\mu, \Sigma)$ having the same first two moments as a *uniform random point inside the rectangle*. This is a moment-matching representation, not a claim that the image pixels are Gaussian.

The center is the mean:

$$\boxed{\mu = \begin{bmatrix} c_x \\ c_y \end{bmatrix}.} \tag{6}$$

For $X \sim U[-w/2, w/2]$, $\mathrm{Var}(X) = w^2/12$. The corresponding axis-aligned covariance is

$$D = \begin{bmatrix} w^2/12 & 0 \\ 0 & h^2/12 \end{bmatrix}. \tag{7}$$

The counterclockwise rotation matrix is

$$R_\alpha = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}. \tag{8}$$

Covariance transforms as $\mathrm{Cov}(AX) = A\,\mathrm{Cov}(X)A^\top$. Therefore,

$$\boxed{\Sigma = R_\alpha D R_\alpha^\top.} \tag{9}$$

Writing $\Sigma = \left[\begin{smallmatrix} a & c \\ c & b \end{smallmatrix}\right]$ gives

$$a = \frac{w^2}{12}\cos^2\alpha + \frac{h^2}{12}\sin^2\alpha, \tag{10}$$

$$b = \frac{w^2}{12}\sin^2\alpha + \frac{h^2}{12}\cos^2\alpha, \tag{11}$$

$$c = \frac{w^2 - h^2}{12}\sin\alpha\cos\alpha. \tag{12}$$

Thus the center controls μ, while size and orientation jointly control Σ.

4 The Bhattacharyya distance

Given two boxes, form Gaussians $P = \mathcal{N}(\mu_1, \Sigma_1)$ and $Q = \mathcal{N}(\mu_2, \Sigma_2)$. The Bhattacharyya coefficient is

$$BC(P, Q) = \int_{\mathbb{R}^2} \sqrt{p(x)q(x)}\,dx, \qquad 0 \le BC \le 1. \tag{13}$$

The corresponding distance is $D_B = -\log BC$. Let $\bar{\Sigma} = (\Sigma_1 + \Sigma_2)/2$ and $\delta = \mu_1 - \mu_2$. For nondegenerate Gaussians,

$$\boxed{D_B = \underbrace{\frac{1}{8}\delta^\top \bar{\Sigma}^{-1}\delta}_{\text{center mismatch}} + \underbrace{\frac{1}{2}\log\frac{\det\bar{\Sigma}}{\sqrt{\det\Sigma_1\det\Sigma_2}}}_{\text{shape/orientation mismatch}}.} \tag{14}$$

The first term penalizes differences in centers, accounting for scale and orientation. The second penalizes differences in covariance, so both shape and angle influence the loss. It is zero for identical distributions.

5 Probabilistic IoU and the box loss

Since $BC = e^{-D_B}$, the Hellinger distance satisfies

$$H(P,Q) = \sqrt{1 - BC(P,Q)} = \sqrt{1 - e^{-D_B}}. \tag{15}$$

A Gaussian-based ProbIoU similarity is then

$$\boxed{\text{ProbIoU}(B,\hat{B}) = 1 - \sqrt{1 - e^{-D_B(B,\hat{B})}}.} \tag{16}$$

The core regression objective is

$$\boxed{\mathcal{L}_{\text{box}} = 1 - \text{ProbIoU}(B,\hat{B}) = \sqrt{1 - e^{-D_B(B,\hat{B})}}.} \tag{17}$$

Numerical implementations clamp or stabilize expressions near degenerate covariance matrices and near perfect overlap. ProbIoU is *not* identical to polygon IoU; rotated polygon IoU remains appropriate for evaluating geometric overlap.

5.1 Why the loss respects 180-degree symmetry

Because $R_{\alpha+\pi} = -R_\alpha$,

$$\Sigma(\alpha+\pi) = (-R_\alpha)D(-R_\alpha)^\top = R_\alpha D R_\alpha^\top = \Sigma(\alpha). \tag{18}$$

Consequently the Gaussian, and thus the ideal ProbIoU box loss, cannot distinguish α from $\alpha+\pi$; nor should it, because they describe the same rectangle. Likewise,

$$\Sigma(w,h,\alpha) = \Sigma(h,w,\alpha+\pi/2). \tag{19}$$

An important subtlety: for $w = h$, $\Sigma = (w^2/12)I$ does not depend on angle. This is compatible with the intuitive difficulty of assigning a stable axis to a square.

6 The complete detector objective

OBB detection requires more than a regression loss. For the standard Ultralytics rotated-bounding-box training path, a schematic objective is

$$\boxed{\mathcal{L}_{\text{OBB}} = \lambda_{\text{box}}\mathcal{L}_{\text{box}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{DFL}}\mathcal{L}_{\text{DFL}}.} \tag{20}$$

Classification: prediction locations receive positive/negative supervision after target assignment. A typical binary cross-entropy term for one class with target t and score s is

$$\ell_{\text{BCE}}(s,t) = -t\log s - (1-t)\log(1-s). \tag{21}$$

DFL: instead of regressing each boundary offset as one scalar, the detection head predicts a categorical distribution over distance bins. If the target distance is $d \in [k, k+1]$, the distribution-focal target interpolates between neighboring bins:

$$\boxed{\ell_{\text{DFL}}(d,p) = -(k+1-d)\log p_k - (d-k)\log p_{k+1}.} \tag{22}$$

The expected-bin value gives a continuous predicted offset. The exact offsets, normalization, and weighting are implementation-specific. The box loss and DFL are computed on assigned foreground predictions; classification includes the relevant positive/negative prediction locations.

Version dependence: Some more recent rotated-detector implementations add explicit angle-related terms. A dedicated angle term is *not* a universal or necessary fourth component of every YOLO11-OBB release. To document one experiment exactly, inspect the installed Ultralytics version, its rotated detection criterion, and the configured loss gains. Do not infer the training objective solely from the number of values predicted by the head.

7 Worked geometric comparison

Let the ground-truth register be $B = (250, 250, 120, 30, 30^\circ)$. Consider predictions

$$\hat{B}_1 = (253, 248, 118, 32, 35^\circ),$$
$$\hat{B}_2 = (250, 250, 120, 30, 210^\circ).$$

The first has center, size, and orientation discrepancies and therefore a positive box loss. The second has exactly the same geometry as the ground truth: $210^\circ = 30^\circ + 180^\circ$, so its Gaussian parameters coincide and its ideal ProbIoU box loss is zero. A naive angle-MSE would incorrectly assign the second prediction a large error.

8 Connection to reading orientation

There are *two separate objectives* in a water-meter pipeline:

$$\boxed{\text{OBB localization} \longrightarrow \text{horizontal rectification} \longrightarrow 0^\circ/180^\circ \text{ semantic correction.}} \tag{23}$$

The OBB loss should be invariant under 180° rotation because the rectangle is geometrically identical. But a digit sequence is not necessarily semantically identical after such a rotation. A downstream orientation classifier must resolve the remaining binary ambiguity using image content or contextual cues. More precise bounding-box training *cannot by itself* teach the network which end of the register is the beginning of the reading.

9 Key takeaways

1. OBB parameterizations are nonunique. A geometrically sensible regression loss should respect π-periodicity and width–height exchange.
2. The Gaussian representation $\mu = (c_x, c_y)^\top$, $\Sigma = R_\alpha \operatorname{diag}(w^2/12, h^2/12) R_\alpha^\top$ unifies location, shape, and angle.
3. Bhattacharyya distance compares Gaussian means and covariance; ProbIoU converts it into a bounded box similarity.
4. Detector training also requires classification and, for relevant YOLO implementations, distribution focal loss and prediction-to-target assignment.
5. Geometric OBB orientation and semantic reading orientation are different problems.

References

- L. Yang et al., *The KFIoU Loss for Rotated Object Detection* (related Gaussian-based geometric losses), arXiv:2201.12558.
- J. Llerena et al., *Gaussian Bounding Boxes and Probabilistic Intersection-over-Union for Object Detection*, arXiv:2106.06072.
- X. Li et al., *Generalized Focal Loss: Learning Qualified and Distributed Bounding Boxes for Dense Object Detection*, NeurIPS 2020.
- Ultralytics, Oriented Bounding Boxes documentation; source code (check the specific installed version).